

prior admissions, demographics, claims, etc.

Data ingestion: In this step, data is moved from source systems into ML platforms like data lakes, warehouses, or feature stores. An example is batch ingestion of nightly ETL jobs from operational databases into a lakehouse.

Data transformation and cleaning: Here, raw and messy data is turned into structured, usable data (such as removing duplicates of the same patient visits recorded twice).

Data analysis: This involves exploratory data analysis (EDA) to understand distributions, relationships, and issues. An example is a correlation analysis between features and the target.

Data visualisation: Visual storytelling makes data patterns and issues understandable to stakeholders – for example, visualising customer churn by tenure and product usage.

Feature engineering: In this step, meaningful input variables (features) are created from raw data. An example is the number of patient admissions in a hospital in the last 12 months.

Model exploration: This involves trying different model families and approaches to see what fits best – for example, LSTMs for time-series and Transformers for text in deep learning.

Model building: The chosen model(s) are formalised into reproducible code and pipelines, such as implementing a training script that reads from the feature store, trains, and logs metrics.

Model training: The training process is run at scale, often multiple times, in this step. An example is training on millions of patient records to predict readmission.

Model evaluation: This assesses how well the model performs and whether it's fit for its purpose – for example, fairness and performance across age, gender, region, or other sensitive groups.

Model deployment: The model is made available to real users and systems in this step. An example

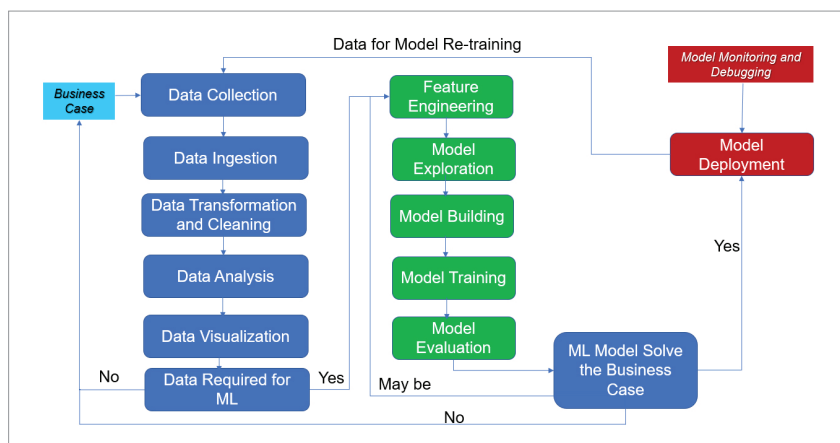


Figure 3: MLOps framework

is a REST endpoint that returns readmission risk when given patient data in API deployment.

Model monitoring and debugging: This involves ongoing oversight of the model in production such as monitoring prediction drift.

Role of an MLOps engineer

MLOps engineers acts as the bridge between development and operations. Their roles focus on:

Foundational DevOps skills:

Managing the operation and deployment aspects of the ML workflow using traditional software engineering practices.

Standardised model development:

Implementing industry best practices to ensure models are reproducible, scalable, and compliant with organisational standards.

Pipeline management: Designing and managing end-to-end pipelines from data ingestion to automated model deployment, so that data scientists can focus on higher-level tasks.

Open source MLOps platforms

A modern MLOps stack combines traditional DevOps tools with specialised machine learning platforms (Table 2).

Here's a brief description of these tools.

Apache NiFi: This visual data-flow tool is designed for moving and transforming data between systems. It supports drag-and-drop pipelines, real-time streaming, and strong data provenance. NiFi excels in regulated environments because every data movement is tracked and auditable.

MLOps framework steps	Tools used
Data ingestion	Apache NiFi, Airbyte, Slinger, Kafka
Orchestration	Apache Airflow, Kubeflow Pipelines
Data processing	Apache Spark, dbt Core, Great Expectations
Analytics and exploration	JupyterLab, Apache Superset
Model development	PyTorch, TensorFlow, Scikit-Learn
CI/CD and packaging	Jenkins, Docker
Model registry	MLflow Model Registry
Observability	Prometheus + Grafana, ELK Stack
Feedback loop	Kafka

Table 2: Open source tools used in the different steps of the MLOps framework